Tianyou Ma

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Research Interests

- **Chromatic methods in symplectic topology**: Kirwan maps and their kernels in Morava K-theory.
- **Mirror symmetry**: Teleman's conjecture on Fukaya categories and Hamiltonian reduction.
- **Quantum cohomology**: quantum Kirwan maps and Gromov–Witten invariants of quotients.

Education

Australian National University · BSc (Honours) in Mathematics. Supervisor: Brett Parker. WAM 84.18/100, GPA 6.73/7 to date (Honours marks incomplete; final results December 2026). Jul 2025 – Dec 2026

Australian National University · BSc in Mathematics. WAM 84.75/100, GPA 6.75/7. Jul 2024 – Jun 2025

Shandong University · BSc in Mathematics (ANU joint pathway). Sep 2021 – Jun 2025

GRE Mathematics Subject Test · 970. Sep 2026

Research

The Integral Cohomology of Smooth Abelian Symplectic Quotients Preprint, Oct 2026
For compact Hamiltonian torus actions that are free on the zero level, we show that the kernel of the integral Kirwan map is generated by classes vanishing on half spaces, as Tolman–Weitsman showed over $\mathbb{Q}$. The proof goes through Morava K-theory. Available on request.

Chromatic Kirwan Theory (ANU Honours thesis) Submitted Oct 2026
Supervised by Brett Parker. Its last chapter grew into the paper above.

Optimization Theory from the Perspective of Information Geometry · undergraduate thesis, Shandong University, supervised by Cheng Zhao. 2025

Conference Talk

The Integral Cohomology of Smooth Abelian Symplectic Quotients · 70th Annual Meeting of the Australian Mathematical Society, Perth. Dec 2026 (planned)

Expository Talks

Chromatic Kirwan Theory · Honours thesis talk, ANU. Nov 2026 (scheduled)

The Atiyah–Guillemin–Sternberg Convexity Theorem for Moment Maps · Lie Groups and Lie Algebras course, ANU. Nov 2025

How to Understand A_∞-Algebras · public mathematics talk, ANU. Oct 2025

The Adams Spectral Sequence · Vector Bundles and K-Theory course, ANU. May 2025

Seminars Organised

Morse Theory and Floer Theory · reading seminar, ANU: from Morse homology to Hamiltonian Floer homology and Kuranishi structures. Co-organiser (with Zizheng Yang) and main speaker. Nov 2025 – Mar 2026

Complex Algebraic Geometry · seminar, ANU: detailed proofs of the Kodaira vanishing and embedding theorems. Organiser and main speaker. Apr–Jun 2026

Schools, Courses and Reading Programmes

SMRI Summer School · Sydney Mathematical Research Institute, University of Sydney (funded). Jan 2027

Bridgeland Stability Conditions · semester-long research reading programme run by Anand Deopurkar and Nicolás Vilches, ANU. 2026

Homological Mirror Symmetry · advanced course by Yixuan Li, ANU: wrapped Fukaya categories, Fukaya–Seidel categories, enumerative mirror symmetry. 2025–26

Supersymmetry, Localisation, and Instanton Counting · special course run by Bailing Wang and Yixuan Li, ANU. Witten's supersymmetric localisation, GIT and its physical interpretation. Sem. 1, 2026

AMSI Summer School, *Algebraic Knot Theory* · Sydney (funded). Jan 2025

Conferences and Workshops

70th Annual Meeting of the Australian Mathematical Society · University of Western Australia, Perth. Dec 2026 (expected)

Online: SLMath workshops on motivic homotopy theory and on geometric representation theory and 3d mirror symmetry; Harvard CMSA workshop on Lagrangian Floer theory. Sep–Oct 2026

Gauge Theory and String Geometry Workshop · ANU. Nov–Dec 2025

Awards

SMRI Summer School Grant · Sydney Mathematical Research Institute. 2026

Outstanding Undergraduate Thesis Award · highest mark in the department (1/80), Shandong University. 2025

AMSI Summer School Grant · travel and accommodation support. 2025

Dean's International Science Excellence Scholarship, ANU College of Science. 2024